Seminar 12 - GR 312 CD - 13.01.2022

Ecuații diferențiale liniare de ord I

$y'(x) = f(x)\,y + g(x)$; f, g cunoscute ; $?\ y(\cdot)$ nec ; $y \in C^1(I)$

Metoda variației constantelor a lui Lagrange.

I) **Etapa omogenă** $y' = f(x)\cdot y \Rightarrow \frac{y'}{y} = f(x)$ ec cu var sep. $\int dx$

$\int \frac{y'(x)}{y(x)}\,dx = \int f(x)\,dx \overset{+C}{\Rightarrow} \ln|y(x)| = \int f(x)\,dx + C \Rightarrow$

$|y(x)| = e^{\int f(x)dx + C} \Rightarrow y(x) = \pm e^{\int f(x)dx}\cdot e^{C}$; $\pm e^C = K \Rightarrow$

$\Rightarrow \boxed{y_o(x) = K\cdot e^{\int f(x)dx}}$ sol omogenă.

II) **Etapa neomogenă** $y' = f(x)\cdot y + g(x)$. (*)

caut sol de tipul $y(x) = K(x)\cdot e^{\int f(x)dx}$ sol pt (*)

$\Rightarrow y' = f(x)\cdot K(x)\cdot e^{\int f(x)dx} + g(x)$

$y' = K'(x)e^{\int f(x)dx} + K(x)\cdot e^{\int f(x)dx}\cdot f(x)$ $\Big\} \Rightarrow g(x) = K'(x)e^{\int f(x)dx}$

$\Rightarrow K'(x) = g(x)e^{-\int f(x)dx} \Rightarrow K(x) = C + \int g(x)e^{-\int f(x)dx}dx.$

$$\boxed{y(x) = e^{\int f(x)dx}\cdot\left[C + \int g(x)e^{-\int f(x)dx}dx\right]}\ ;\ C \in \mathbb{R}.$$

∞ de sol.

Obs: Dc. $y(x_0) = y_0 \Rightarrow$ C se determină în mod unic $\Rightarrow$ sol unică.

Ex 1 $y'(x) + \frac{1}{2}y = \frac{1}{2}e^{x/3}$.

$y'(x) = \underbrace{-\frac{1}{2}}_{f(x)}\cdot y + \underbrace{\frac{1}{2}e^{x/3}}_{g(x)}$ $\quad f(x) = -\frac{1}{2}$. $\quad g(x) = \frac{1}{2}e^{x/3}$.

Met I: a) Etapa omogenă: $y' = -\frac{1}{2}y \Rightarrow \frac{y'}{y} = -\frac{1}{2} \mid \int dx \Rightarrow$

$\Rightarrow \int \frac{y'(x)}{y(x)}dx = -\int \frac{1}{2}dx \Rightarrow \ln|y(x)| = -\frac{x}{2} + C \Rightarrow |y(x)| = e^{-\frac{x}{2}}\cdot e^C$.

$\Rightarrow \boxed{y_o(x) = K\cdot e^{-\frac{x}{2}}}$ sol omog.

b) Etapa neomogenă. $y' = -\frac{1}{2}y + \frac{1}{2}e^{x/3}$ (*)

$y(x) = k(x)\cdot e^{-\frac{x}{2}}$ sol pt (*) $\Rightarrow y' = -\frac{1}{2}k(x)e^{-\frac{x}{2}} + \frac{1}{2}e^{x/3}$

$\Rightarrow y' = k'(x)e^{-\frac{x}{2}} + k(x)\cdot e^{-\frac{x}{2}}\cdot(-\frac{1}{2})$] $\Rightarrow$

$\Rightarrow \frac{1}{2}e^{\frac{x}{3}} = k'(x)e^{-\frac{x}{2}} \Rightarrow k'(x) = \frac{1}{2}e^{\frac{x}{3}}\cdot e^{\frac{x}{2}} \Rightarrow k'(x) = \frac{1}{2}e^{\frac{x}{3}+\frac{x}{2}} = \frac{1}{2}e^{\frac{5x}{6}}$.

$\Rightarrow k(x) = \int \frac{1}{2}e^{5x/6}dx = \frac{1}{2}\frac{e^{5x/6}}{5/6} + C = \frac{3}{5}e^{5x/6} + C$.

$y(x) = \left(C + \frac{3}{5}e^{5x/6}\right)\cdot e^{-\frac{x}{2}} \Rightarrow \boxed{y(x) = C\cdot e^{-\frac{x}{2}} + \frac{3}{5}e^{x/3}}$.

$y_0(x)$ sol. omog. $y_p(x)$ sol partic la ec neomog

$\boxed{y(x) = y_0(x) + y_p(x)}$

Met II. $\boxed{y' = -\frac{1}{2}\cdot y + \frac{1}{2}e^{x/3}}$

$f(x)$ $g(x)$.

$$y(x) = e^{\int f(x)dx}\cdot\left[C + \int g(x)e^{-\int f(x)dx}dx\right] =$$

$$= e^{-\int \frac{1}{2}dx}\left[C + \int \frac{1}{2}e^{x/3}e^{\int \frac{1}{2}dx}dx\right] =$$

$$= e^{-\frac{x}{2}}\left[C + \int \frac{1}{2}e^{x/3}\cdot e^{x/2}dx\right] = e^{-\frac{x}{2}}\left[C + \frac{1}{2}\int e^{\frac{5x}{6}}dx\right] =$$

$$= e^{-\frac{x}{2}}\cdot C + \frac{1}{2}e^{-\frac{x}{2}}\cdot\frac{e^{5x/6}}{5/6_3} = C\cdot e^{-\frac{x}{2}} + \frac{3}{5}e^{x/3}.$$

Met III. $\begin{cases} g(x) = A\cdot e^{\alpha x} \\ y' = f(x)\cdot y + g(x) \end{cases} \Rightarrow$

a) $y' = f(x)\cdot y \Rightarrow y_0(x)$ sol omog.

b) $g(x) = Ae^{\alpha x} \Rightarrow$ căutăm sol partic pt ec. neomog y_p. o forție de forma $y_p = B\cdot e^{\alpha x}$.

a) $y_0(x) = C\cdot e^{-x/2}$.

b) $g(x) = \frac{1}{2}e^{x/3} \Rightarrow ?\ y_p(x) = Be^{x/3}$ sol pt (*). $\Rightarrow$

$\Rightarrow y_p' = B/3\cdot e^{x/3} = -\frac{1}{2}Be^{x/3} + \frac{1}{2}e^{x/3} \Rightarrow \frac{B}{3} + \frac{B}{2} = \frac{1}{2} \Rightarrow \frac{5B}{6} = \frac{1}{2} \Rightarrow B = \frac{3}{5}$.

$\Rightarrow \boxed{y_p(x) = \frac{3}{5}e^{x/3}} \Rightarrow y(x) = y_0(x) + y_p(x) = Ce^{-x/2} + \frac{3}{5}e^{x/3}$. ☐

Ex 2 · $y' = 2y + 4 - t$ · (*)

Met I: a) $y' = 2y \Rightarrow \int \frac{y'}{y} = \int 2 \Rightarrow \ln|y| = 2t + C \Rightarrow |y| = e^{2t+C} \Rightarrow$

$\Rightarrow y_0(t) = Ke^{2t}$.

b) $y(t) = K(t)e^{2t}$ sol pt (*) $\Rightarrow y' = 2K(t)e^{2t} + 4 - t \Rightarrow$

$\Rightarrow y' = K'(t)e^{2t} + K(t) \cdot 2e^{2t} \Rightarrow$

$4 - t = K'(t)e^{2t} \Rightarrow K'(t) = e^{-2t}(4-t) \Rightarrow$

$\Rightarrow K(t) = \int (4-t)e^{-2t}dt = 4\int e^{-2t} - \int te^{-2t} = 4 \cdot \frac{e^{-2t}}{-2} -$

$- \int t \cdot \left(\frac{e^{-2t}}{-2}\right)' dt = -2e^{-2t} - \left[t \cdot \frac{e^{-2t}}{-2} - \int \frac{e^{-2t}}{-2} dt\right] =$

$= -2e^{-2t} + \frac{t}{2}e^{-2t} + \frac{1}{2}\frac{e^{-2t}}{-2} + C = -2e^{-2t} + \frac{t}{2}e^{-2t} + \frac{1}{4}e^{-2t} + C$.

$= e^{-2t}\left(\frac{4}{-2} + \frac{1}{4} + \frac{t}{2}\right) + C = e^{-2t}\left(-\frac{7}{4}\right) + \frac{t}{2}e^{-2t} + C$.

$y(t) = e^{2t}\left[e^{-2t}\left(-\frac{7}{4} + \frac{t}{2}\right) + C\right] \Rightarrow y(t) = -\frac{7}{4} + \frac{t}{2} + Ce^{2t}$.

$y_p(t)$ $y_0(t)$

Verificare. $y_p = -\frac{7}{4} + \frac{t}{2}$.

$y_p' = \frac{1}{2}$; $\Leftrightarrow y_p' = 2y + 4 - t = 2\left(-\frac{7}{4} + \frac{t}{2}\right) + 4 - t =$

$= -\frac{7}{2} + 4 + t - t = +\frac{1}{2}$.

Met III: ? $y_p(t) = at + b \Rightarrow y_p' = a$; $y_p' = 2(at+b) + 4 - t$.

$g(t) = 4 - t$.
polinom de ord I.

$a = 2(at+b) + 4 - t$; $\forall t$.

$\begin{cases} a = 2b + 4 \\ 2a - 1 = 0 \Rightarrow a = \frac{1}{2} \end{cases} \Rightarrow b = \frac{a-4}{2} = \frac{\frac{1}{2}-4}{2} = -\frac{7}{4}$ $\Rightarrow$

$\Rightarrow y_p(t) = \frac{1}{2}t - \frac{7}{4} \Rightarrow y(t) = y_0(t) + y_p(t) = ke^{2t} + \frac{t}{2} - \frac{7}{4}$

Met II: $y(t) = e^{\int f(t)dt}\left[C + \int g(t)e^{-\int f(t)dt}dt\right] = \ldots$

Ex 3 a) $tx' + 2x = 4t^2$; $x(1) = 2$. ? $x(t)$.

b) $x' + x = 5\sin 2t$.

c) $(1+t^2)x' + 4tx = (1+t^2)^{-2}$.

d) $tx' + 2x = \sin t$; $t > 0$.

e) $tx' - x = t^2 e^{-t}$; $t > 0$.

f) $2x' + x = 3t^2$.

g) $y' - y = 2te^{2t}$; $y(0) = 1$.

h) $ty' + 2y = t^2 - t + 1$; $y(1) = \frac{1}{2}$; $t > 0$.

i) $y' - 2y = e^{2t}$; $y(0) = 2$.

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b) $x' + x = 5\sin 2t$. $\Rightarrow x_0(t)$

$g(t) \Rightarrow \begin{cases} x_p(t) = A\cos 2t + B\sin 2t \\ x_p' = \dots \end{cases}$? $A = -2$. $B = -1$.

Sisteme liniare omogene cu coef const

$$\begin{cases} x'(t) = a\cdot x(t) + b\cdot y(t) \\ y'(t) = c\cdot x(t) + d\cdot y(t) \end{cases}$$

a, b, c, d cunoscute.

$x(\cdot)$; $y(\cdot)$ = ?

$\Leftrightarrow$ reducerea sist la o ec. dif de ord II.

$\Leftrightarrow$ metoda valorilor și vectorilor proprii.

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix}; \quad \vec{\varphi}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}; \quad \vec{\varphi}'(t) = \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}.$$

$$\boxed{\vec{\varphi}'(t) = A\cdot\vec{\varphi}(t)} \quad (*)$$

. ? x i. $x' = a\cdot x \Rightarrow \frac{x'}{x} = a \Rightarrow \int \frac{x'}{x} = \int a \Rightarrow \ln|x| = at + C \Rightarrow \boxed{x = e^{at}\cdot k}$

$\vec{\varphi}(t) = e^{\lambda t}\cdot\vec{v}$ sol pt $(*) \Rightarrow \vec{\varphi}' = A\cdot e^{\lambda t}\vec{v}$; $\vec{\varphi}' = \lambda e^{\lambda t}\vec{v}$ $\Rightarrow \boxed{A\vec{v} = \lambda\cdot\vec{v}}$.

$\Rightarrow \lambda$ val proprie pt A

$\vec{v}$ vect pr asoc lui λ